

DBMS LAB EXPERIMENT 1 (Modified for Classroom Allotment System)

AIM

To Create, Alter, Drop, Rename and Truncate tables for the Classroom Allotment System using Data Definition Language (DDL) statements.

DESCRIPTION

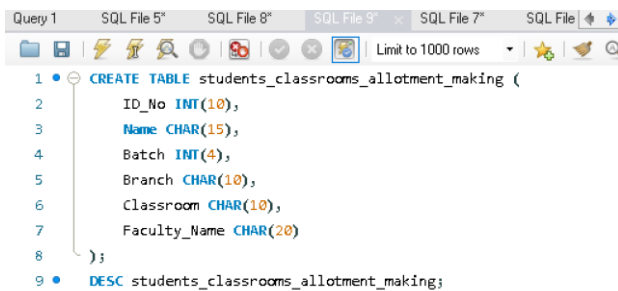
DDL (Data Definition Language) statements are used to define and modify the database structure.

Commands: CREATE, ALTER, DROP, TRUNCATE and RENAME.

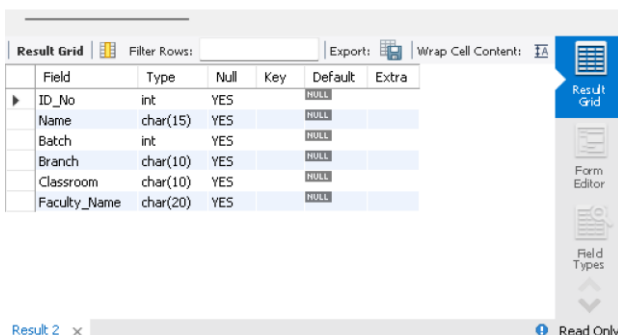
QUESTIONS

1) Create a table STUDENT

```
CREATE TABLE student(  
ID_No INT,  
Name VARCHAR(20),  
Batch INT,  
Branch VARCHAR(10),  
Fee INT,  
Grade CHAR(2)  
);
```



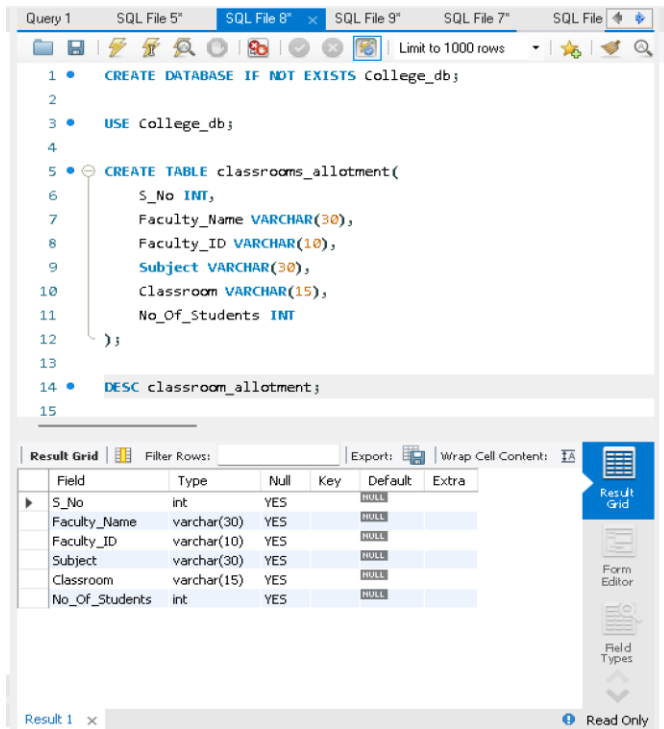
```
1 • CREATE TABLE students_classrooms_allotment_making (  
2     ID_No INT(10),  
3     Name CHAR(15),  
4     Batch INT(4),  
5     Branch CHAR(10),  
6     Classroom CHAR(10),  
7     Faculty_Name CHAR(20)  
8 );  
9 • DESC students_classrooms_allotment_making;
```



Field	Type	Null	Key	Default	Extra
ID_No	int	YES		NULL	
Name	char(15)	YES		NULL	
Batch	int	YES		NULL	
Branch	char(10)	YES		NULL	
Classroom	char(10)	YES		NULL	
Faculty_Name	char(20)	YES		NULL	

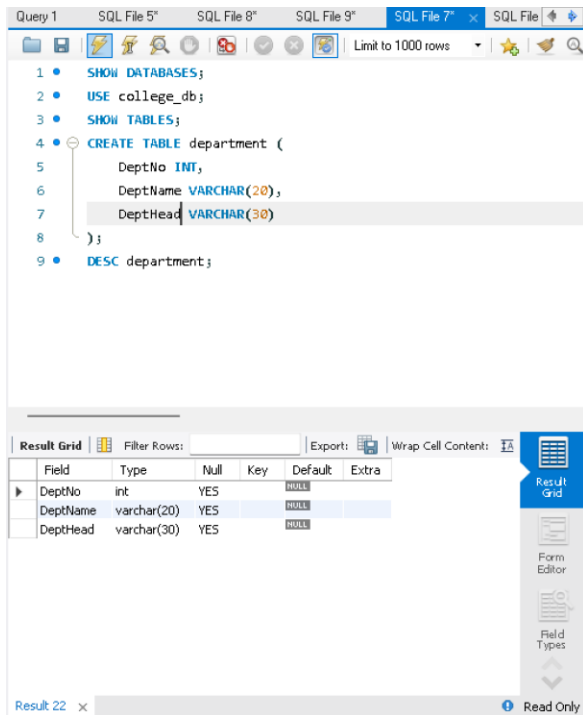
2) Create a table FACULTY (Classroom Allotment)

```
CREATE TABLE faculty(  
FacultyID VARCHAR(10),  
FacultyName VARCHAR(25),  
Subject VARCHAR(25),  
Classroom VARCHAR(10),  
NoOfStudents INT  
);
```



3) Create a table DEPARTMENT

```
CREATE TABLE department(  
DeptNo INT,  
DeptName VARCHAR(20),  
DeptHead VARCHAR(25)  
);
```



4) Create a table COURSE

```

CREATE TABLE course(
CourseNo VARCHAR(5),
CourseName VARCHAR(30),
CourseType VARCHAR(20),
SemNo INT,
HallNo VARCHAR(10),
FacNo VARCHAR(10)
);

```

The screenshot shows a SQL IDE with multiple tabs. The active tab, 'SQL File 8', contains the following SQL code:

```

1 • SHOW DATABASES;
2 • USE classroom_allotment;
3 • CREATE TABLE course (
4     CourseID INT,
5     CourseName VARCHAR(30),
6     Classroom VARCHAR(15)
7 );
8 • DESC course;

```

Below the code editor, the 'Result Grid' is displayed, showing the structure of the 'course' table. The grid has columns for Field, Type, Null, Key, Default, and Extra.

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(5)	YES		NULL	
CourseName	varchar(25)	YES		NULL	
NoOfStudents	int	YES		NULL	
Classroom	varchar(10)	YES		NULL	
FacultyID	varchar(10)	YES		NULL	
Branch	varchar(10)	YES		NULL	
Credits	int	YES		NULL	

On the right side of the IDE, there are buttons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom, it says 'Result 1' and 'Read Only'.

5) Modify the FACULTY table by adding Branch

ALTER TABLE faculty

ADD Branch VARCHAR(10);

OUTPUTS

- 1) DESC student;
- 2) DESC faculty;
- 3) DESC department;
- 4) DESC course;
- 5) DESC faculty;

```

1 • USE college_db;
2 • DROP TABLE course;
3 • CREATE TABLE course
4 • (
5 •     CourseNo VARCHAR(5),
6 •     CourseName VARCHAR(25),
7 •     NoOfStudents INT,
8 •     Classroom VARCHAR(10),
9 •     FacultyID VARCHAR(10),
10 •     Branch VARCHAR(10)
11 • );ALTER TABLE course
12 ADD Credits INT;
13 • DESC course;

```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(5)	YES		NULL	
CourseName	varchar(25)	YES		NULL	
NoOfStudents	int	YES		NULL	
Classroom	varchar(10)	YES		NULL	
FacultyID	varchar(10)	YES		NULL	
Branch	varchar(10)	YES		NULL	
Credits	int	YES		NULL	

RESULT

The STUDENT, FACULTY, DEPARTMENT and COURSE tables for the Classroom Allotment System were successfully created and modified using DDL commands.